GG-K-508b
MARCH 27, 1957
SUPERSEDING
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June 11, 1956 and
Fed. Spec. GG–K–508
July 23, 1946

FEDERAL SPECIFICATION

KNIVES, DENTAL: CRAFTSMAN'S, COMPOUND, AND PLASTER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers knives for use in dental offices and laboratories for trimming compound and plaster.

1.2 Classification.

1.2.1 *Types and styles.*—Dental knives in accordance with this specification shall be of the following types and styles, as specified in the invitation for bids:

Type I.—Knife, craftsman's
  Style A.—Adjustable blade
  Style B.—Fixed blade
Type II.—Knife, office and laboratory
  Style A.—Two-piece handle
  Style B.—One-piece handle
Type III.—Knife, plaster

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specification and standard.—The following specification and standard, of the issues in effect on date of invitation for bids, forms a part of this specification.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–P–12902 — Packaging of Surgical, Dental and Veterinary Instruments, Diagnostic Equipment and Related Items.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Steel.*—The steel for blades shall be as specified hereinafter for each type.

FSC Classes 5110, 6520

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3.1.2 *Wood.*—The wood for handles shall be rosewood, cocobolo, or other suitable hardwood.

3.2 **Cutting edge.** — The blades shall be honed to a keen cutting edge and shall show no undue wear, such as turning over or the breaking of any part, when tested as described in 4.3.2.

3.3 **Marking.** — The manufacturer's name or registered trade-mark shall be neatly and permanently impressed on the blade of each knife.

3.4 Type I, craftsman's knife.

3.4.1 *Design.*—The knife shall conform to the applicable style illustrated in figure 1.

3.4.2 *Dimensions.* — The approximate dimensions shall be as shown in table I.

3.4.3 *Blade.*—The blade shall be made of high-grade cutlery steel, having a Rockwell hardness of C–59 to C–61.

3.4.3.1 *Style A.*—The blade for style A knife shall have a straight cutting edge with back slanting at least 0.75 inch to a bevel tip. The blade shall be easily and securely adjustable to handle to give an operating surface of from 1.50 to 3.50 inches long.

3.4.3.2 *Style B.* — The blade for style B knife shall have a straight cutting edge, with back rounded at least 0.50 inch to a point. The tang shall be securely and permanently attached to the handle.

3.4.4 *Handle.*

3.4.4.1 *Style A.*—The handle for a style A knife shall be suitable for use with extension blades.

3.4.4.2 *Style B.* — The handle for style B knife shall be in one piece, forced on the blade extension and securely attached by means of a steel rivet, which shall be flush with the handle surface.

3.4.5 *Ferrule.*—The ferrule shall be heavily nickel-plated.

3.5 Type II, office and laboratory knife.

3.5.1 *Design.*—The knife shall conform to the applicable style illustrated in figure 1.

3.5.2 *Dimensions.* — The approximate dimensions shall be as shown in table II.

3.5.3 *Blade.*—The blade shall be made of steel having a carbon content of approximately 0.85 percent, with a Rockwell hardness of not less than C–52. The blade shall have a straight cutting edge; shall be rigid throughout; and shall have a heavy rib or back rounded at least 0.75 inch to a point for style A and at least 0.50 inch for style B. Style A shall have a tang extending the full length of the handle.

3.5.4 *Handle.*

3.5.4.1 *Style A.*—The handle for style A knife shall be in two pieces, securely attached to the blade extension by means of not less

TABLE I.—*Type I, craftsman's knife*

Style	Length over-all	Length of handle	Diameter of handle (at widest part)	Length of blade	Width of blade	Thickness of blade
	Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Inch*
A	(adjustable	4.25	0.937	4.75	0.281	0.031
B	5.25	3.75	0.875	(bevel tip) 1.50	0.281	0.062

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TABLE II.—*Type II, office and laboratory knife*

Style	Length over-all	Length of handle	Length of blade	Width of blade	Thickness of blade, at back near handle
	Inches	*Inches*	*Inches*	*Inch*	*Inch*
A	6.00	3.75	2.00	0.375	0.093
B	6.68	[1]4.68	1.87	0.375	0.109

[1] Length includes ferrule.

than three steel rivets, which shall be flush with the handle surface.

3.5.4.2 *Style B.*—The handle for style B knife shall be in one piece, forced on the blade extension and equipped with a heavily nickel-plated ferrule.

3.6 Type III, plaster knife.

3.6.1 *Design.*—The knife shall be of the type known to the dental trade as Gritman plaster knife, of the design shown in figure 1, suitable for trimming plaster and separating flasks.

3.6.2 *Dimensions.* — The approximate dimensions shall be as shown in table III.

TABLE III.—*Type III, plaster knife*

Length over-all	Length of handle	Length of blade	Width of blade	Thickness of blade through handle
Inches	*Inches*	*Inches*	*Inch*	*Inch*
6.50	4.00	2.12	0.75	0.125

3.6.3 *Blade.*—The blade shall be made of steel having a content of 1.00 to 1.10 percent carbon and 0.22 to 0.25 percent manganese, with a Rockwell hardness of C–58 to C–62. The blade shall have a straight cutting edge properly ground and free from burs. The end shall be rounded 0.62-inch to a point. The tang shall extend the full length of the handle and approximately 0.50 inch beyond, tapering to a width of approximately 0.50 inch and to a thickness of approximately 0.03 inch, forming a chisel end.

3.6.4 *Handle.*—The handle shall be in two pieces riveted to the tang with not less than three evenly spaced rivets of brass or other suitable metal. The handle shall be furnished flush with the blade side edges.

3.7 Workmanship. — The workmanship shall be first class in every respect. The knives shall be symmetrically made, smoothly and properly finished, and sharpened for immediate use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—One knife of each type delivered shall be taken at random for inspection and tests.

4.2 Inspection.—Samples shall be carefully inspected for material, workmanship, design, dimensions, marking, and finish.

4.3 Tests.

4.3.1 *Test for hardness.*—The hardness of the blades shall be determined on a Rockwell-hardness testing machine.

4.3.2 *Test for cutting edge.*—Each knife shall be tested by cutting at least ten shavings from a strip of oak or other hardwood.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing (see Federal Standard No. 102).

5.1.1 *Unit of issue.*—One knife, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units indicated in table IV, shall be packaged in each unit, intermediate, and exterior container as applicable for the required level of protection specified in the procurement document.

TABLE IV.—*Packaging quantities*

Federal Stock No.	Unit package	Intermediate package	Shipping container
5110–537–2000	1	12	288
6520–537–3000	1	12	360

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Procedures.*—Packing and packaging shall be in accordance with Military Specification MIL–P–12902 for the applicable procedure code number indicated in table V.

TABLE V.—*Procedures*

Level of protection	Procedure code No.
Level A	38
Level B	22

5.1.4 *Level C.* — The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.—Marking shall be in accordance with Military Specification MIL–P–12902, except that ASMPA stock number is not required.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data. — Purchasers should specify type and style required and should exercise any desired options offered herein. (See 1.2.1, 5.1, 5.1.3, 5.2.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.5 Transportation description.—Transportation description applicable to these items is:

Cutlery not otherwise indexed by name
Carload minimum weight 20,000 pounds subject to Rule 34 Uniform Freight Classification
Motor Volume minimum weight 20,000 pounds

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6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

5110–537–2000 Knife, Craftsman's Dental
6520–537–3000 Knife, Plaster, Dental

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

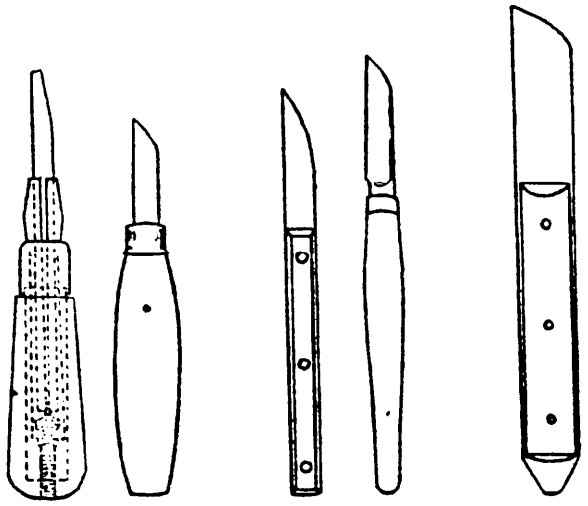


FIGURE 1.—Knives, dental.

Copies of this specification may be purchased for 10 cents each.

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